INCH POUND

MIL-DTL-5041L (USAF)
23 September 2019

SUPERSEDING
MIL-PRF-5041K
20 February 2007

DETAIL SPECIFICATION

TIRES, RIBBED TREAD, PNEUMATIC, AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

AMSC: N/A

FSC: 2620

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FOREWORD

1. This detail specification is approved for use by the AF70 Hill AF Base, AFMC 417 SCMS/GUEA Landing Gear/Aircraft Tires Program Office Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

2. This specification covers requirements for aircraft pneumatic tube type and tubeless ribbed tread tires intended for use on military aircraft wheels.

3. Comments, suggestions, or questions on this document should be addressed to Landing Gear/Aircraft Tires Program Office, 417th SCMS, 6040 Gum Lane, Bldg 1216, Hill AFB, UT 84056-5825 or 748SCMG.Workflow@us.af.mil attn: 417th Landing Gear/Aircraft Tires Program Office. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database https://assist.dla.mil.

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SUMMARY OF CHANGES

PAGE	MODIFICATIONS
Page I – Title	Changed
Page ii – Forward	Added
Page iii – Summary of changes	Added
Page iv – vii Table of contents	Added

PARAGRAPH/SECTION	MODIFICATIONS
2.1	Changed
2.2.1	Changed
2.2.2 – 2.2.3	Changed
2.3	Changed
3.1 – 3.2	Changed
3.2.1 – 3.2.2	Changed
3.3	Changed
3.4 – 3.4.2	Changed
3.5.1	Changed
3.5.3 – 3.5.3.2	Changed
3.5.4 – 3.5.4.1.2	Changed
3.5.5 – 3.5.5.2	Changed
3.5.6 – 3.5.8	Changed
3.5.9 – 3.5.9.5	Changed
3.5.10 – 3.5.12	Changed
3.6 – 3.6.4	Changed
3.7 – 3.8	Changed
4.1 a – b	Changed
4.2	Changed
4.3 – 4.3.2 d	Changed
4.4 – 4.5	Changed
4.6 – 4.6.6.3	Changed
4.7	Changed
5.1	Changed
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6.2 – 6.8	Changed
Table I – Table III	Changed
Figure 1	Changed
Figure 2	Added
Appendix A (entire)	Added
Concluding Material	Changed

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1. SCOPE

1.1 Scope. This specification covers requirements for aircraft pneumatic tube type and tubeless ribbed tread tires intended for use on military aircraft wheels.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections three (3) and four (4) of this specification. This section does ***not*** include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections three (3) and four (4) of this specification, whether or ***not*** they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS AND STANDARDS

SD-6	Standardization Document – Provisions Governing Qualification - Qualified Products Lists and Qualified Manufacturers Lists
MIL-STD-129	Military Marking for Shipment and Storage
MS14113	Tape, Identification, Color Coded, for Aircraft Tires
QPL-5041	Tires, Ribbed Tread, Pneumatic, Aircraft
Other Specifications	Weapon-system specific – see Table I

(Copies of these documents are available online at https://assist.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Drawings	Weapon-system specific – see Table I

(Copies of these drawings are available at https://www.jedmics.net/index.jsp, https://assist.dla.mil, or contact Landing Gear/Aircraft Tires Program Office, 417th SCMS, 6040 Gum Lane, Bldg 1216, Hill AFB, UT 84056-5825, or email at 748SCMG.Workflow@us.af.mil attn: 417th Landing Gear/Aircraft Tires Program Office)

FEDERAL AVIATION ADMINISTRATION PUBLICATION

TSO-C62	Technical Standard Order – Aircraft Tires

(Copy of this document is available online at https://www.faa.gov/.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AS4833	Aircraft New Tire Standard – Bias and Radial
SAE ARP6404	Aircraft Tire Electrical Resistance Test

(Copies of these documents are available online at www.sae.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Copy of this document is available online at www.asq.org/.)

AMERICAN SOCIETY FOR TESTING & MATERIALS (ASTM)

ASTM D746	Standard Test Methods for Brittleness Temperature of Plastics and Elastomers by Impact
ASTM D413	Standard Test Methods for Rubber Property – Adhesion to Flexible Substrate

(Copies of these documents are available online at www.astm.org.)

THE TIRE AND RIM ASSOCIATION, Inc. (TRA)

Aircraft Year Book	TRA Standards and related information approved by the Associate for tires, rims and allied parts for aircraft

(Copy of this document is available online at www.us-tra.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern. Some specification sheets are listed in Table I, but the list is ***not*** exhaustive.

3.1.1 Tire and Rim Association (TRA) Aircraft Yearbook. In the event that no specification sheet exists, the TRA Aircraft Yearbook shall be the authority for tire dimensions.

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3.2 Qualification. The tires furnished under this specification shall be products authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3) in accordance with Standardization Document (SD)-6. Manufacturer shall notify the cognizant engineering authority when there are changes in plants, construction, materials, or processes to determine if the change will affect performance of the tire. If performance is affected, the cognizant engineering authority will determine re-qualification requirements. Commercial source retread tires used in military application may ***not*** comply with specific requirements cited herein and must be authorized by the cognizant engineering authority.

3.3 General requirements. Unless otherwise specified, tires shall be suitable for use on military aircraft, on all types of improved and unimproved runways, including ships (usually aircraft carrier decks), under all weather conditions. For tires ***not*** found in Table I, the procuring activity will provide the performance and interface requirements.

3.3.1 Ambient temperature range. The ambient temperature range, which a tire is required to operate, is: -58 degrees (°) to 125° F (-50° to 52° C). All tire compounds shall maintain their fracture toughness and strength characteristic with this temperature range.

3.3.2 Tire sizing and classification. Tire shall be sized and classified in accordance with either the TRA Aircraft Yearbook or the applicable specification sheet.

3.3.3 Retreadability. Re-treadable tires shall be designed and manufactured so that the casing structure, inner liner, sidewall, beads, and bead seat cover shall be satisfactory for repeated retreading.

3.4 Materials. Material quality shall meet performance requirements in the applicable specification sheet. The use of toxic chemicals, hazardous substances, and ozone depleting chemicals shall be avoided, whenever possible. All aircraft tires shall be constructed of materials that meet electrical resistance requirements in accordance with SAE ARP6404.

3.5 Performance.

3.5.1 Basic tire performance. Unless otherwise specified, the basic performance of the tire shall be as specified in the applicable specification sheet.

3.5.2 Tire speeds. Unless otherwise specified, the tire shall have a minimum velocity capability of 120 MPH at the rated load and rated inflation pressure.

3.5.3 Burst pressure. The tire shall be designed to withstand a minimum burst pressure as specified in the applicable specification sheet. If burst pressure is ***not*** specified it shall be 3.5 times the rated pressure for land-based aircraft, and four (4) times rated pressure for carrier-based aircraft.

3.5.4 Dynamic durability. Unless otherwise specified, excluding wear, tires for fighter, attack, and trainer aircraft shall withstand a minimum of 50 cycles of taxi, takeoff, and landing; tires for other aircraft shall withstand a minimum of 100 cycles of taxi, takeoff, and landing. In addition, the Department of Navy aircraft with carrier-based requirements, tires shall withstand normal catapults, landing arrestments, and cable strikes without failure.

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3.5.4.1 Cord fraying. During the dynamic life of the tire, cord fraying, if present in the groove of the tire, shall be only of the outer layer or cord.

3.5.4.2 Tread chunking. During the dynamic life of the tire, the tread chunk shall ***not*** exceed one (1) square inch in area or 75% (percent) of the skid; there shall be no more than three (3) chunks, each ½ to one (1) inch in area; and there shall be no more than 10 chunks totaling more than four (4) square inches in area. There shall be no rib undercutting.

3.5.4.3 Groove cracking. During the dynamic life of the tire, there shall be no groove cracking in tires having rubber or reinforced fabric tread. In tires with fabric tread, any void in the bottom of the groove shall be no deeper than a void caused by the outer layer of cord being pulled through the rubber stock in bottom of the groove. There shall be no rib undercutting.

3.5.4.4 Bead separation. During dynamic life of the tire, the walls of the bead bundle shall ***not*** show evidence of separating from the casing plies. If bead wires are used, individual wires shall ***not*** show evidence of separating from each other, or kinked, broken, or exposed.

3.5.4.5 Cable bruise – carrier-based tires only. A tire to be used on carrier-based aircraft shall withstand the minimum cable bruise load specified in the applicable specification sheet without pressure loss.

3.5.5 Wheel/tire slippage. A mounted tire inflated to rated inflation pressure shall ***not*** slip on the rim to such an extent that would damage the tube or valve in a tube type tire or the inflation seal of a tubeless tire.

3.5.6 Inflation pressure retention – tubeless tire. After an initial 12-hour stretch starting at rated inflation pressure, the pressure loss from rated inflation pressure in a tubeless tire assembly during the subsequent 24-hour period shall ***not*** exceed five (5) % of rated inflation pressure specified in the applicable specification sheet. The tire shall ***not*** show evidence of appearance defects such as sidewall blisters, tread separation, etc.

3.5.7 Balance. Tires shall be balanced, when ***not*** inflated, within tolerances specified in the applicable specification sheet. Out-of-tolerance conditions may be corrected by utilizing balance pads affixed to the inside of the tire. In tube type tires, the pads shall ***not*** chafe the tubes. Balance pad adhesion values shall be as follows: eight (8) pound force per inch – width minimum for tubeless tires and 1.5 pounds force per inch-width minimum for tube type tires.

3.6 Design and construction. Design and construction shall conform to the requirements specified in the applicable specification sheet.

3.6.1 Tire dimensions and weight. Dimensions and weight of the tire shall be as specified in the applicable specification sheet. All lettering and decorative ribs and designs shall be included in the dimensions. Dimension definitions are found in the TRA Aircraft Yearbook or the specification sheet.

3.6.2 Tire dimensions for helicopters. Helicopter tire dimensions shall be in accordance with the applicable specification sheet.

3.6.3 Tread. Tire tread may be manufactured using fabric-reinforced tread, fabric tread, or rubber.

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3.6.3.1 Tread pattern. The tread shall be a ribbed tread pattern having a minimum of three (3) grooves for tires with a cross-sectional width greater than 6.0 inches, and minimum of two (2) grooves for tires with a cross-sectional width of 6.0 inches or less. The grooves shall be continuous and circumferential.

3.6.3.2 Underskid thickness. The material between bottom of the groove and first fabric layer shall be of sufficient thickness to minimize groove cracking. This requirement does ***not*** apply to fabric tread tires.

3.6.3.3 Maximum Wear Limit (MWL)indicator. If a colored indicator is used, it shall be visible when observing a cross-sectional cut.

3.6.3.4 Retread Buff Line (RBL). On tires which are to be retreaded, RBL shall be a definitive, continuous, circumferential layer of uninterrupted rubber 0.060 inch minimum thickness extending shoulder to shoulder. The RBL shall be located adjacent to and above the carcass or breaker plies in the finished tire.

3.6.4 Sidewall. The sidewall shall protect the casing against abrasion and weathering.

3.6.5 Venting. Means shall be provided to vent any trapped gases in the sidewall. If vent holes are used, there shall be at least eight (8) functioning vent holes per sidewall located above the rim flange.

3.6.5.1 Tube type tires. If vent holes are used, they shall be marked with a white or aluminum colored dot. Vent holes may penetrate the liner of the tire.

3.6.5.2 Tubeless tires. If vent holes are used, they shall be marked with a bright green dot. Vent holes shall ***not*** penetrate the liner of the tire.

3.6.6 Rim dimensions. Tires shall be designed and constructed to fit the rims as specified in the applicable specification sheet. If no specification sheet exists, the rim shall conform to the specification control drawing or to the rim profile recommended by the TRA Aircraft Yearbook.

3.6.7 Cut limit dimension. The cut limit is the allowable depth for cuts as measured from the bottom of the nearest groove in the vicinity of the cut. The cut limit shall be determined by the manufacturer. It shall be determined by calculating the number of structural cords that can be cut to allow continued safe operation of the tire.

3.6.8 Bead.

3.6.8.1 Bead seating pressure. A bead seating pressure within the range specified in Table II shall be required to accomplish the initial seating of the tire bead on the rim ledge of a wheel having contour conforming to the applicable specification sheet.. This shall be achieved without using a lubricant on either bead or rims.

3.6.8.2 Chafing resistance. The bead shall be protected to prevent chafing of the tire in the rim area under normal operating conditions. The protecting material in tubeless tires shall ***not*** leak air or nitrogen.

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3.6.8.3 Trimming. The bead toes of all tires used on tube type applications shall be inspected for excessive toe flash or sharp edges which could cut or chafe the tube. If trimming the bead toe results in a sharp edge, the edge shall be buffed to a smooth contour. Casing plies shall ***not*** be cut during trimming or buffing.

3.7 Product identification and marking. Unless otherwise specified, the following information, as applicable, shall be marked legibly and permanently on the tire sidewall for the life of the casing. Markings on the tire sidewall shall be located so that they can ***not*** be removed during buffing for re-treading. See Figure 1.

3.7.1 Date of manufacture and serial number. Each tire shall be marked with a unique serial number. The serial number shall consist of a maximum of 10-digits. It is recommended that the first five (5) digits shall be the date of manufacture in the form of a Julian date. The remaining digits, ***not*** to exceed five (5), selected by the manufacturer, may be numbers, letters, or a combination thereof. If the date is ***not*** included in the serial number, the date of manufacture (either Julian or month/year – MM/YY) shall be marked legibly and permanently on the sidewall of the tire.

3.7.2 Manufacturer's name or trademark. Include one or the other, or may include both.

3.7.3 Maximum Wear Limit (MWL). The MWL marking shall be either "MWL-RC" for tires with a colored indicator or "MWL #" for tires with a ply count indicator to be shown in a separate molded box (see Figure 1 and Figure 2). For tires with an outside diameter greater than 26-inches, the MWL marking shall be molded in a minimum of two (2) places equally spaced on each sidewall of the tire. For tires with an outside diameter less than or equal to 26-inches, the MWL marking shall be molded once on each sidewall, 180 degrees from the cut limit marking, as shown in Figure two (2). The lettering should be a minimum of 3/8 inch high and the dimensions of the rectangle should be a minimum of ½ inch by 1¼ inch in height. Tires for the Department of Navy shall ***not*** incorporate MWL marking.

3.7.4 Size.

3.7.5 Tube type tires. Include "TUBE TYPE."

3.7.6 Ply rating. Ply Rating or PR is permissible.

3.7.7 Cut-limit dimension identification. For tires with an outside diameter greater than 26-inches, the cut-limit dimension identification shall be molded in a minimum of two (2) places equally spaced on each sidewall of the tire. For tires with an outside diameter less than or equal to 26-inches, the cut-limit dimension identification shall be molded once on each sidewall, 180° from the cut-limit marking on the opposite sidewall, as shown in Figure one (1). The cut-limit dimension should be expressed as increments of 1/32 inch and should be rounded to the next smaller increment of 1/32 inch when a fraction of 1/32 inch is involved. The lettering should be a minimum of ¼ inch in height and the diameter of the circle should be a minimum of one (1) inch.

3.7.8 National Stock Number (NSN). The NSN shall be molded on one side of the tire, on the same side as the serial number. The prefix NSN shall be included. The NSN shall ***not*** contain dashes or spaces (Example: NSN 2620XXXXXXXXX). The height of the letters and the numbers shall ***not*** be less than ¼ inch.

3.7.9 Manufacturer's Qualification Test (QTR). QTR number, prefixed by "QTR."

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3.7.10 Fabric tread tires. Tires with fabric tread shall be marked: “FABRIC TREAD”.

3.7.11 Country of manufacture. If other than USA.

3.7.12 Manufacturer’s mold number.

3.7.13 Manufacturer’s part number.

3.7.14 Color dots for vent holes. If used.

3.7.15 Additional markings. As required by the applicable specification sheet.

3.7.16 Markings not required for military tires with equivalent commercial applications. The following markings are ***not*** required: MWL, cut limit dimension, NSN, and QTR.

3.8 Interchangeability. All parts having the same manufacturer’s part number shall be functionally and dimensionally interchangeable.

3.9 Age. The tire shall ***not*** be more than 36-months old from the date of manufacture to the initial date of delivery.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

4.1.1 Qualification inspection. (see 4.3)

4.1.2 Conformance inspection. (see 4.4)

4.2 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.3 Qualification Inspection. Qualification inspections shall be performed on tires when qualification is required (see 3.2 and 6.3). Testing shall include all the tests specified in paragraph 4.7. The qualification test samples shall consist of at least one (1) tire of each construction, size, and type. The tire(s) shall be identified and marked as specified in paragraph 3.7. A waiver of the marking requirements on tires supplied for qualification tests may be granted when agreed to between the procuring activity and the Contractor.

4.3.1 Qualification Test Report (QTR). The manufacturer shall prepare test reports in the general format shown in Appendix A, and shall furnish electronically to the Government approving activity.

4.3.1.1 Bead-seating. If bead-seating pressure tests are conducted on steel or non-aircraft wheels, this shall be specified in the QTR.

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4.3.1.2 Load Deflection curves. The QTR shall be accompanied by a chart showing load-deflection curves. The deflection shall be plotted on the y-axis, and the load on the x-axis.

4.3.1.2.1. There shall be seven (7) inflation pressure curves on the chart. One (1) of these curves shall be rated inflation pressure. Three (3) curves shall be less than the rated inflation pressure curve at approximately 85%, 70%, and 55% of the rated inflation pressure. Three (3) of the curves shall be greater than the rated inflation pressure at approximately 115%, 130%, and 160% of the rated inflation pressure. For Navy tires, a load deflection curve at carrier inflation pressure shall be included.

4.3.1.2.2 All inflation pressure curves shown on the chart shall have a range from 16% deflection to the tire bottoming point.

4.3.1.2.3 The following percent deflection lines shall be plotted across the inflation pressure curves based on dimensions at rated inflation pressure: 16, 20, 24, 28, 32, 35, 37, 40, 44, and 48.

4.3.1.2.4 The following data shall be listed in the lower right-hand corner of the load deflection chart:

Size_____ PR
Manufacturer
QTR Number
Rated Static Load (lb) _____ at _____ psi
Rated Speed (mph)
Rim Dia (in) _____Flange Ht (in)_____
Flat Tire Radius (in) _____
Data at Rated Static Load or Inflation (or both)
OD (in) _____ Section width (in)
Shoulder Dia (in)
Shoulder Width (in)
Loaded Radius (in)
Contact Area (sq in)
Bottomed Pressure (psi)
Bottomed Load (lb)

4.3.1.3 Footprints. Images indicating the actual shape, total gross contact area, and net contact area in square inches at rated static load and inflation, shall be submitted electronically with each QTR. The prints shall include data at rated load, at increments of approximately 15%, 30%, and 45% above rated load, and at increments of approximately 15%, 30%, and 60% below rated load. For tires used by the Navy on carrier-based aircraft, a footprint at bottomed condition at carrier inflation pressure shall also be submitted. Exceptions to this requirement due to safety reasons shall be noted in the QTR.

4.3.1.3.1 Tire size.

4.3.1.3.2 Ply rating.

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4.3.1.3.3 Rated load and inflation.

4.3.1.3.4 Net and gross contact area.

4.3.1.3.5 Date.

4.3.1.3.6 Manufacturer's name.

4.3.1.3.7 QTR number.

4.3.1.4 Navy QTR Inflated profiles. Profiles indicating actual inflated shape of the tire at rated and carrier inflation pressures shall be submitted with each Navy QTR.

4.3.1.5 Cross section image. The supplier shall submit an 8 by 10-inch digital photograph taken perpendicular to and showing clearly the complete cross section of the tire with adequate contrast between the ends and body of the carcass, breaker and tread reinforcement cords, and the remainder of the tire section. The tire section shall be photographed with the beads spread so that the distance from heel-to-heel is the same as the flange-to-flange distance for the tire. A steel tape or ruler shall be laid at the base of the beads of the tire to show the distance between the bead heals. The photographs shall be submitted along with the QTR to the Government approving activity.

4.3.2 Qualification tests. The qualification tests shall consist of all the tests specified in paragraph 4.7. Qualification approval shall apply to a specific construction, size, and materials as defined in the manufacturer's QTR. The construction shall be identified by the manufacturer's QTR number.

4.3.3 Dynamic Test Inspection. When the tires are dynamically tested, a section of the tire shall be submitted to the Government approving activity to determine any evidence of failure. The section shall be approximately two (2) inches in width and representative of the worst area. When the approving activity tests the tire, that activity reserves the right to section the tire in any manner necessary for the examination. The manufacturer shall submit an image taken perpendicular to and showing clearly the complete cross-section of the tire with adequate contrast between the ends and body of the casing, breaker, and tread reinforcement cords and the remainder of the tire section. The beads of the cross-section shall be spread so that the distance from heel-to-heel is the same as the flange-to-flange distance for the tire. A steel tape or ruler shall be laid at the base of the beads of the tire to show the distance between the bead heals. The images shall be submitted along with the QTR to the Government approving activity.

4.4 Conformance Inspection. Conformance inspection shall include individual tests and sample tests.

4.4.1 Individual tests. Each tire shall be subject to the following tests:

4.4.1.1 Examination of product (see 4.7.1)

4.4.1.2 Balance check – when limits are specified (see 4.7.2)

4.4.2 Sample tests. Unless otherwise specified, tires shall be sampled in accordance with ASQ/ANSI Z1.4 at an initial inspection level or normal and the following tests shall be performed:

4.4.2.1 Tire dimensions, weight, and rim interface (see 4.7.3)

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4.4.2.2 Balance pad adhesion (see 4.7.4)

4.4.2.3 Inflation pressure retention for tubeless tires (see 4.7.8)

4.4.2.4 Burst pressure (see 4.7.9)

4.5 Test conditions. Unless otherwise specified in the individual test description, all tests shall be conducted at ambient temperature and pressure as specified in applicable paragraphs below.

4.6 Requirement cross-reference matrix. Table III provides a cross-reference matrix of the Section three (3) requirements tested or verified in the paragraphs below.

4.7 Test methods.

4.7.1 Examination of product test. The tire and associated documentation shall be examined to determine compliance with size, material, dimensions, weight, bead width, tread pattern, sidewall, venting, identification, interchangeability, and age requirements.

4.7.2 Balance test. The tire shall be balance checked to determine that the moment required to static balance the tire does ***not*** exceed the limits specified in the applicable specification sheet.

4.7.3 Tire dimensions, weight, and rim interface test. The tires shall be mounted on its rim, inflated to rated inflation pressure specified in the applicable specification sheet, allowed to stand for 12-hours minimum at room temperature, and then re-adjusted to rated inflation pressure. The tire dimensions and weight shall then be determined and compared to the values in applicable specification sheet. Tire weight may be measured un-mounted. For radial tires, the inflated dimensions may be used rather than the grown dimension to verify proper dimensions. Manufacturers shall ensure the new tire inflated dimensions are properly correlated to the grown dimensions to ensure conformance to the grown tire envelope.

4.7.4 Balance pad adhesion test. Balance pad adhesion shall be tested in accordance with ASTM D413, to determine compliance with the requirements specified in paragraph 3.5.7.

4.7.5 Bead seating pressure test. The bead seating pressure can be measured by employing an electrical contact system to determine when the bead has been seated against the wheel flange. After placement of shim stock at a minimum of three (3) locations at 120 degrees apart around the tire, the tire is inflated. The pressure which the last shim makes electrical contact can be considered the bead seating pressure. The test shall be accomplished without using lubricant on either bead or rims. The pressure shall be within the range specified in Table II.

4.7.6 Dynamic durability test. A dynamic durability test shall demonstrate satisfactory tire performance during the taxi, takeoff, and landing cycles. Unless otherwise specified, test parameters for taxi, takeoff, and landing are specified in the applicable specification sheet. A suggested test method for 120 MPH, entitled: *Alternate Qualification Procedures – 120 MPH Rated Tires* is provided in SAE AS4833.

4.7.6.1 Dynamic durability test temperature. Unless otherwise specified, the contained air temperature or highest casing temperature at the start of 80 percent of the taxi, takeoff, and landing cycles shall be no less than 105° ± 5° F (41° ± 3° C).

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4.7.6.2 Cable bruise test (carrier based aircraft tires only). Unless otherwise specified, the tire shall be inflated to carrier pressure, and vertically loaded against a 1.625-inch diameter length of plain round steel stock or equivalent diameter arresting gear cable to the specified vertical load in the applicable specification sheet. Unless otherwise specified, immediately following the release of this load, the tire shall be subjected to the same loading conditions 180 degrees from the application of the first load.

4.7.6.3 Post-test inspection. At the conclusion of the dynamic durability test, the examination of product test specified in paragraph 4.7.1, shall be performed with the tire at rated inflation pressure. In addition, the tire and associated documentation shall be examined to determine compliance with the fabric material, chafing resistance, MWL indicator (if applicable), tread chunking, groove cracking, cord fraying, bead separation, cable bruise (for carrier-based aircraft tires), RBL, and wheel/tire slippage requirements. The post-test inspection shall include examination of a cross-sectional cut.

4.7.6.4 Bead separation test. If bead bundle or wire separation is found in the cross-sectional cut, the material around the bead bundle shall be stripped back at least one inch to determine if the separation was caused by the cross-sectional cut of the post-test inspection or during the dynamic durability test. If no separation is found in the stripped area, the bead construction shall be considered satisfactory.

4.7.7 Low temperature test. Samples of the tread, casing, and liner compounds shall be tested in accordance with ASTM D746 at -58° F (-50° C). An alternate test method may be utilized provided that data is submitted substantiating an equivalent test method and the alternate method is approved by the responsible procurement activity.

4.7.8 Inflation pressure retention test – for tubeless tires. The tire shall be inflated to rated inflation pressure specified in the applicable specification sheet and allowed to stand for a minimum of 12-hours, at which time the pressure loss due to stretch shall be replaced. The tire shall then stand for an additional 24-hours, at which time the pressure shall be recorded. Ambient temperature shall be measured at the start and finish of the test to assure that any pressure change was ***not*** caused by ambient temperature change. At no time shall the tire be inflated above rated inflation pressure.

4.7.9 Burst pressure test. The minimum burst pressure specified in the applicable specification sheet shall be applied to the mounted tire and held to a minimum of three (3) seconds. The tire shall ***not*** fail under this pressure. The burst pressure test of a tubeless tire may be conducted with a tube in the same manner as for tube type tires. The tire pressure shall then be increased until the tire bursts or the limit of the equipment is reached, provided the limit of the equipment is over the minimum burst pressure. The testing agency shall report whether failure was a casing or a bead failure.

4.7.10 Conductive material test. All aircraft tires shall be constructed of materials that meet electrical resistance requirements in accordance with SAE ARP6404.

5. PACKAGING

5.1 General. For acquisition purposes, the packaging requirements shall be specified in the contract or purchase order. When actual packaging of materiel is to be performed by DoD personnel, these personnel will contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements will be maintained by the Inventory Control Point packaging activity with the Military Department or Defense Agency, or within the Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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5.2 Marking for Shipment. Each tire shall be labeled in accordance with MIL-STD-129. The identification shall consist of the following information listed in the order shown:

5.2.1 National Stock Number (NSN). NSN: Tire, Pneumatic, Aircraft Specification MIL-DTL-5041

5.2.2 Color coding, age identification. The tire shall be color-code taped for age identification using the color of tape in accordance with MS14113. Color used shall reflect year of casing manufacture, with the date of year superimposed on the tape.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Tires covered by this specification are intended for use on all military aircraft.

6.2 Acquisition requirements. Acquisition documents specifically include, but are not limited to the following, unless specified in this specification:

6.2.1 Title, number, and date of this specification.

6.2.2 The referenced applicable documents.

6.2.3 Operational environment.

6.2.4 Performance or interface requirements.

6.2.5 Basic tire performance requirements.

6.2.6 Retreadability requirements.

6.2.7 Taxi, takeoff, and landing cycle conditions to be met.

6.2.8 Tire identification data.

6.2.9 Initial inspection level or sample tests.

6.2.10 Test conditions.

6.2.11 Dynamic durability test requirements.

6.2.12 Packaging requirements.

6.2.13 Data required.

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6.3 Qualified Parts List (QPL). For products requiring qualification, manufacturers are urged to study the requirements of this document. Contract award will only be made on products qualified to be included in the QPL5041, whether or not the products have been specified by contract award date. For manufacturers to be eligible for contract award or purchase orders for products covered by this specification, they are urged to coordinate qualification testing with the Government for their products. Product qualification information or qualification test reports may be obtained from Landing Gear/Aircraft Tires Program Office, 417th SCMS, 6040 Gum Lane, Bldg. 1216, Hill AFB, UT 84056-5825 or 748SCMG.Workflow@us.af.mil attn: 417th Landing Gear/Aircraft Tires Program Office (see 3.1). Refer to the SD-6 for product qualification procedures. And, visit the Qualified Products Database at https://qpd.dla.mil/qpd/.

6.4 Additional Information.

6.4.1 Use of tubeless tire in tube type application. In tube type applications, tubeless tires (with a tube installed) may be used in lieu of tube type tires.

6.5 Definitions.

6.5.1 Bead bundle. High tensile strength steel wires embedded in rubber. The beads anchor the casing plies and provide a firm-mounting surface on the wheel.

6.5.2 Carrier pressure. Unless otherwise specified in applicable standards or drawings in Table I for testing purposes, carrier pressure is a minimum of 1.3 times the rated inflation pressure.

6.5.3 Casing. The structural part of the tire.

6.5.4 Casing ply. One (1) or more strips or layers of fabric found in the casing.

6.5.5 Fabric-reinforced tread. A fabric reinforced tread is at one with a fabric ply or plies constructed in the material between the outer casing ply and the bottom of the tread grooves.

6.5.6 Fabric tread. A fabric tread is at one with a fabric ply or plies constructed in the tread ribs above the bottom of the tread grooves.

6.5.7 Julian date. The Julian date is a five (5) digit alphanumeric code indicating the year and number of the day of the year. As an example: the Julian date for May 23, 1996 – the number of the day is 143 of 1996, it would appear as 96143. An acceptable variation of the Julian date using a decade identifier may be used and assigned by the manufacturer. This variation is also a five (5) digit alphanumeric code indicating the year, the number of the day of the year, and a decade identifier. For example, the Julian date May 23, 1996 would state the number of the day as 143 of 1996, and would appear as 6143J, “J” being the decade identifier.

6.5.8 Liner. The inner layer of low permeability rubber in a tubeless tire, which acts as a built-in tube and prevents gas from seeping through the casing plies.

6.5.9 Ply Rating. The term ply rating (PR) is used to identify a given tire with its maximum recommended load when used in a specific type of service. It is an index of the tire strength and does ***not*** necessarily represent the number of cord plies. The ply rating is ***not*** to be interpreted as limiting the number of plies that may be used in a given tire construction.

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6.5.10 Rated inflation pressure. The rated inflation pressure is the specified inflation pressure corresponding to the rated load for the tire.

6.5.11 Rated load. Rated load is the maximum permissible load at the specified inflation pressure. Rated loads are established and standardized by Tire and Rim Association (TRA) or a specification sheet. The rated load combined with the rated inflation pressure is utilized when selecting tires for application to an aircraft and for testing to the performance requirements of this document.

6.5.12 Retread Buff Line (RBL). The RBL is a definitive, continuous, circumferential layer of uninterrupted rubber, 0.060 inch minimum thick, extending shoulder to shoulder. The RBL is located adjacent to and above the carcass or breaker plies in the finished tire.

6.5.13 Ribbed Tread. A ribbed-type tread pattern is one having continuous circumferential ribs and grooves.

6.5.14 Rubber tread. A rubber tread is at one with no reinforcing plies in the tread material.

6.5.15 Skid depth. Skid depth is the radial distance, measured along the centerline of the tire mold, from the line enveloping the outer cross section of the tread to the line enveloping the outer cross section of the under tread of the deepest groove.

6.6 Subject term (key word) listing.

Bead seating pressure
Chunking tubeless
Cut-limit dimension
Fabric reinforced tread
Fabric tread
Ply rating
Rated inflation pressure
Rated load
Skid depth tread
Tread chunking
Tube type
Tubeless

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes. A Summary of Changes was added, (see page iii).

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TABLE I.
Tire drawing matrix.

WHEEL TYPE 1/	SIZE	PLY RATING	DRAWING or MS NUMBER 2/
TL	8.50 – 10	10	54C763
TL	8.50 – 10	12	3/
TL	12.50 – 16	12	64F1880
TL	20.00 – 20	26	65D1542
TL	18 X 4.4	6	56D1172
TL	18 X 5.5	14	MS26535
TL	20 X 4.4	10	MS26538
TL	20 X 4.4	10	MS26538
TL	20 X 4.4	12	56D1171
TL	20 X 4.4	14	8631427
TL	20 X 5.5	12	MS26540
TL	20 X 5.5	14	MS26540
TL	20 X 5.5	8	MS26539
TT	24 X 5.5	12	MS26526
TL	24 X 5.5	12	MS26526
TT	24 X 5.5	14	MS26526
TL	24 X 5.5	14	MS26526
TT	24 X 5.5	16	MS18060
TT	24 X 7.7	10	MS26558
TL	24 X 7.7	14	58D510
TT	25 X 6.0	16	MS26543
TL	26 X 6.6	14	60C4280
TL	28 X 7.7	14	MS17838
TT	32 X 8.8	18	MS26537
TL	36 X 11	22	8631526
TL	36 X 11	24	MS14482
TL	36 X 11	28	MS90346
TL	38 X 11	14	61D3069
TL	39 X 13	16	63D30091
TL	40 X 14	26	MS26563
TL	44 X 13	26	MS26557
TL	44 X 16	28	61F4307
TL	46 X 16	28	TS0C62

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TABLE I.
Tire drawing matrix – Continued.

TL	49 X 17	26	60D2561
TL	49 X 17	26	71203
TL	56 X 16	38	60D510
TL	13 X 6.0 – 4	14	MS14224(AS)
TL	18 X 5.7 – 8	14	MS14196
TL	18 X 5.7 – 8	18	GD16Vl027
TL	18 X 5.7 – 8	20	GD16VL036
TL	18 X 6.5 – 8	12	63J4242
TL	21 X 7.25 – 10	20	72318
TL	22 X 6.6 – 10	18	8412568
TL	22 X 6.6 – 10	22	MS14168A 74B431200 Rev B 2GLZ36002, Rev C
TL	22 X 6.75 – 10	18	MS14161
TL	22 X 7.75R9	22	PS68-4500068
TL	22 X 7.75 – 9	26	PS68-450068
TL	22 X 8.5 – 11	16	63J4241
TL	23.5 X 7.5R10	N/A	5PTA3065
TL	24 X 6.5 – 14	18	MS14178
TL	24 X 8.0 – 13	18	73453
TL	25.5 X 8.0 – 14	20	GD16VL028
TL	25.5 X 8.0 – 14	20	9235498
TL	26 X 7.75R13	10	MS14483
TL	26 X 7.75 – 13	10	MS14225
TL	26 X 8.75R11	16	MS14223
TL	26 X 10.0 – 11	10	3/
TL	26 X 10.0 – 11	12	3/
TL	27.75 X 8.75 – 14.5	24	GD16VL032
TL	27.75 X8.75R14.5	24	GD16VL032
TL	28 X 9.0 – 14	22	74201
TL	30 X 11.50 – 14.50	26	MS14171
TL	30 X 11.5 – 14.5	26	MS21781
TL	32x11.50 – 15	26	PS-7400055
TL	34.5 X 9.75 – 18	26	8412569
TL	34 X 14.0 – 12	24	DAA3221P508
TL	35 X 11.5 – 16	22	L194C2003

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TABLE I.
Tire drawing matrix – Continued.

TL	36 X 11.0 – 18	20	PS68-410065
TL	36 X 11.0R18	N/A	PS68-410065
TL	37 X 11.5R18	N/A	5PTA3066
TL	40 X 16 – 14	26	PS17026
TL	43 X 16.0 – 20	28	DAA3222P508
TL	B46 X 16.0 – 23.5	30	L194C2025
TL	47 X 18.0 – 18	30	69E177
TL	50 X 21.0 – 20	30	PS17046

Key:
1/ TL Tubeless
1/ TT Tube type
2/ Drawings for Air Force tires contain only performance data and do not contain detail drawings
3/ MS number has not been identified by the Dept of Navy

TABLE II.
Bead seating pressures.

Rated Inflation Pressure (psi)	Minimum Bead Seating Pressure (psi)	Maximum Bead Seating Pressure (psi)
40 or less	25	40
40 to 100	25	*
Over 100	50	*

NOTE – The maximum bead seating pressure shall ***not*** exceed the rated inflation pressure or 200 psi, whichever is less.

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TABLE III.
Paragraph cross-reference matrix.

Section 3 – Requirements Paragraphs	Section 4 – Qualification Test Method
3.2 Qualification	4.7
3.3 General requirements	4.7
3.3.1 Ambient temperature range	4.5, 4.7.6.1, and 4.7.7
3.3.2 Tire sizing and classification	4.7.1
3.3.3 Retreadability	4.7.6.3
3.4 Materials	4.7.1 and 4.7.6.3
3.5.1 Basic tire performance	4.7
3.5.2 Tire speeds	4.7.6
3.5.3 Burst pressure	4.7.9
3.5.4 Dynamic durability	4.7.6
3.5.5 Wheel/tire slippage	4.7.6.3
3.5.6 Inflation pressure retention – tubeless tire	4.7.8
3.5.7 Balance	4.7.2 and 4.7.4
3.6.1 Tire dimensions and weight	4.7.1
3.6.2 Tire dimensions for helicopters	4.7.1 and 4.7.3
3.6.3.1 Tread pattern	4.7.1
3.6.3.2 Underskid thickness	4.7.6.3
3.6.3.3 Maximum wear limit (MWL) indicator	4.7.6.3
3.6.3.4 Retread Buff Line (RBL)	4.7.6.3
3.6.4 Sidewall	4.7.1
3.6.5 Venting	4.7.1
3.6.6 Rim dimensions	4.7.3
3.6.7 Cut limit dimensions	1/
3.6.8.1 Bead seating pressure	4.7.5
3.6.8.2 Chafing resistance	4.7.6.3
3.6.8.3 Trimming	4.7.1
3.7 Product identification & marking	4.7.1
3.8 Interchangeability	4.7.1
3.9 Age	4.7.1

Key:
1/ Determined by manufacturer – no reportable test requirement

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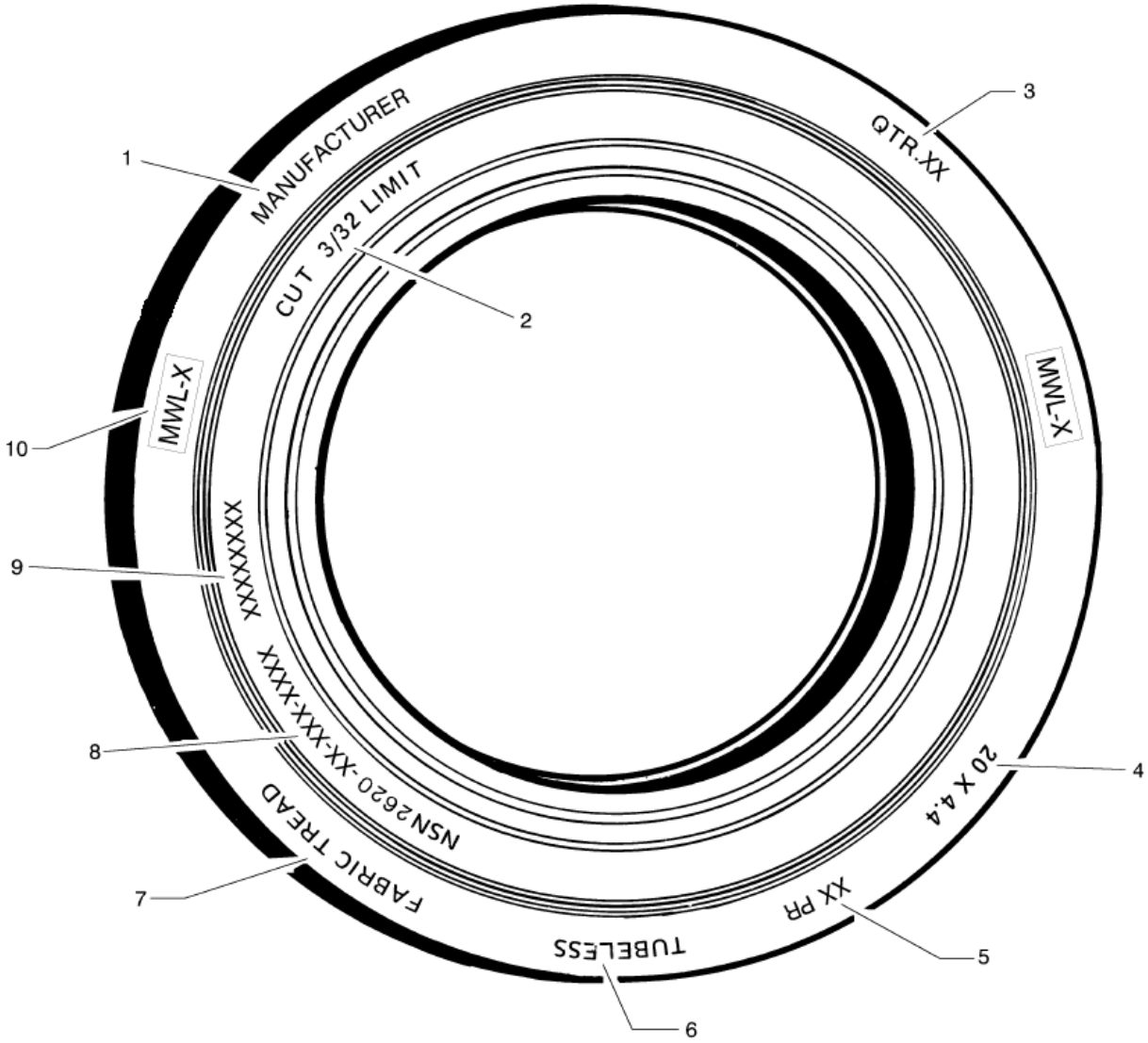


FIGURE 1. Part marking.

Key:

1. Manufacturer’s name or trademark, or both
2. Cut-limit dimension identification
3. Qualification Test Report (QTR) number
4. Size
5. Ply Rating
6. Type (Tube Type or Tubeless)
7. Tread Type
8. National Stock Number (NSN)
9. Date of manufacture and serial number
10. Maximum Wear Limit (if applicable)

NOTE – Exact location is ***not*** required.

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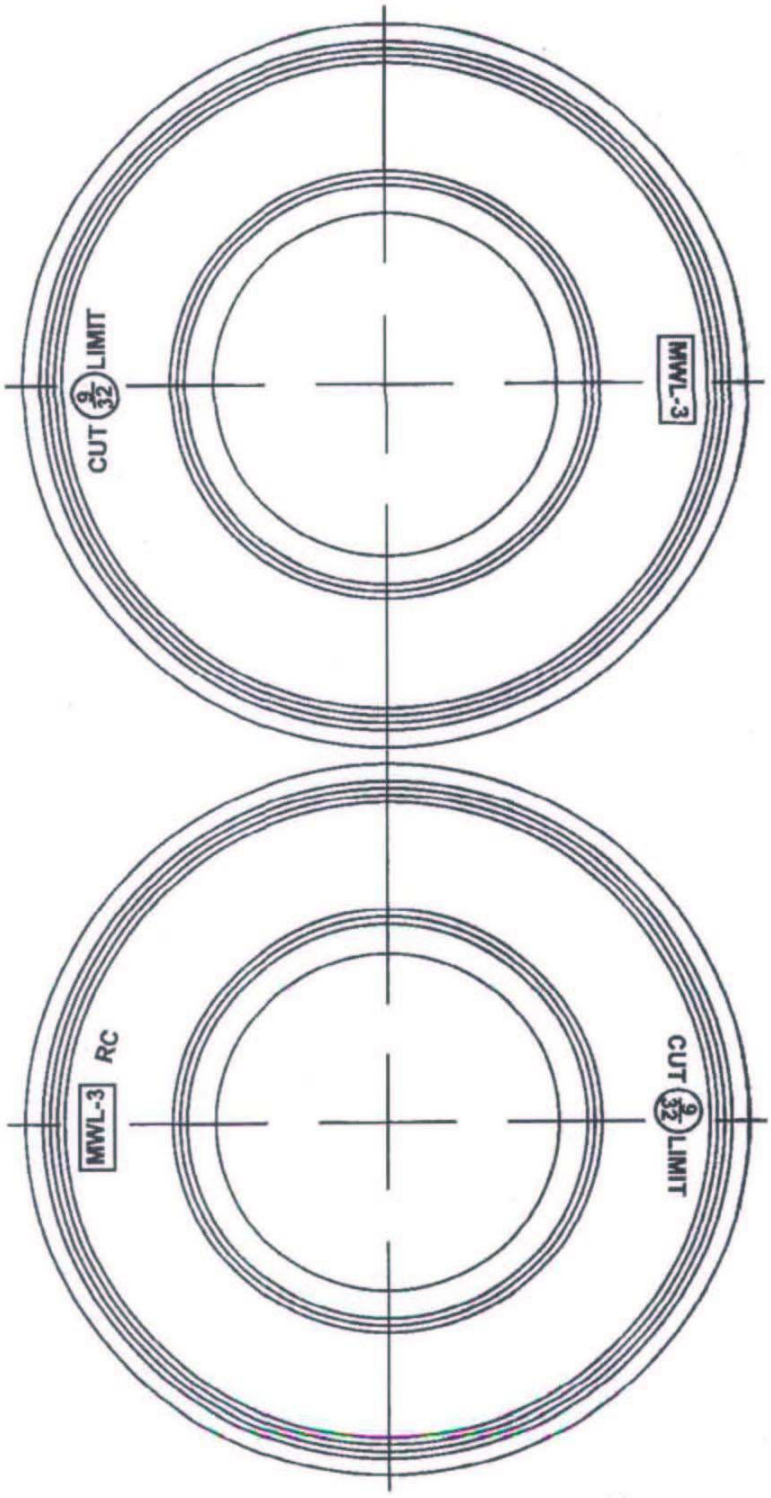


FIGURE 2. Cut-limit dimensions and maximum wear limit identification.

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QUALIFICATION TEST REPORT (QTR) FORMAT

A1. HEADER SECTION

The header for each page, (except the title page) shall be as follows:

[Manufacturer] QTR:
DATE:
PAGE:

A2. TITLE PAGE SECTION

[Company Name] [Location]

[Date]

Qualification Test Report #_____

To

[Customer]

For [Weapon System]

[New or Rebuilt, Bias or Radial, NLG or MLG Tire] [Size]

[NSN]

[P/N (if other than QTR#)]

References: [OEM/military specification], MIL-DTL-5041

Prepared by: _____
Design Engineer

Reviewed by: _____
Design Engineering

Approved by: _____
R&D, QA

Reviewed by: _____
Government Representative

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A3. TABLE OF CONTENTS SECTION.

TABLE OF CONTENTS

ITEM PAGE

QUALITY TEST REPORT TITLE

COMPLIANCE

STATIC TEST REQUIREMENTS / RESULTS
- Measurements
- Load Deflection Curves
  - Static
  - Dynamic / Impact (if applicable)
  - Impact Energy (if applicable)
  - Contact Area (if applicable)
- Tire Contact Areas
- Inflated Tire Profile
- [Other static data required by specification]

DYNAMIC TEST REQUIREMENTS

DYNAMIC TEST RESULTS
- Tire #1
- Tire #2
- Tire #3
- Grown Profile (if applicable)
- Thrown and Grown Profile (if applicable
- [Other dynamic data required by specification]

OTHER PERTINENT DATA
- Torsion Recovery at Low Temperature
- Low Temperature Brittleness Qualification
- High Temperature Compliance (if applicable)
- Quality Control
- Product Identification and Marking
- Photograph
- Cut Limit
- Colored Wear Indicator (if applicable)
- [Other data required by specification]

APPENDICES
- APPENDIX A. FOOTPRINTS
- APPENDIX B. LOAD-SPEED-TIME CURVES
- APPENDIX C. LOW TEMPERATURE APPROVALS
- APPENDIX D. PHOTOGRAPH

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A4. COMPLIANCE SECTION.

COMPLIANCE
The [specific aircraft, N/MLG tire, size] has successfully completed the requirements of [OEM/military specification, revision no., revision date] and MIL-DTL-5041. Test results were obtained on tires manufactured on production equipment at [facility/location] and are representative of future production tires. [Include any general comments concerning test results or post-test condition of tires.]

A5. STATIC TEST REQUIREMENTS / RESULTS SECTION.

STATIC TEST REQUIREMENTS / RESULTS		
Size_____	No. carcass plies_____	Cord material_____
Ply rating_____		
Low temperature approval:		
A) Tread / Carcass: [Identification, Approving Authority, Approval Date]		
B) Inner lining: [Identification, Approving Authority, Approval Date]		
Speed rating_____knots	Rated load_____lbs	Rated pressure_____psi

ITEM		**REQUIREMENTS**	**QUALIFICATION RESULTS**
(A)	Balance	Max _____ in-oz	_____ in-oz
(B)	Bead Seat Pressure	Max _____ psi	_____ psi
		Min _____ psi	_____ psi
(C)	Bead Width	Max _____ in	_____ in
(D)	Burst Pressure	Min _____ psi	_____ psi
(E)	Loaded Radius	Nominal _____ in	_____ in
	Tolerance @ Rated Load And Pressure: _____ to _____ in		
(F)	Dimensions		
	Outside Diameter	Max ____ in	_____ in
		Min ____ in	
	Section Width	Max ____ in	_____ in
		Min ____ in	
	Shoulder Diameter	Max ____ in	_____ in
		Min ____ in	
	Shoulder Width	Max ____ in	_____ in
		Min ____ in	

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A.5. STATIC TEST REQUIREMENTS / RESULTS SECTION – Continued.

ITEM		REQUIREMENTS	QUALIFICATION RESULTS
(G)	Mold Skid Depth	Min_____ in	_____ in
(H)	Total Tread Thickness	Min _____ in	_____ in
(I)	Weight	Max _____ lbs	_____ lbs
(J)	Retread buff line (if applicable)	Min 0.06 inch	_____ in
(K)	Air retention (tubeless)	Max loss 5%	_____ %
	@ Rated inflation pressure and constant ambient temperature		
(L)	Number of wires per bead bundle	(as determined)	_____ 1st
		(as determined	_____ 2nd
		(as determined	_____ 3rd
		(as determined	_____ 4th
(M)	Chafing strips	(as determined)	_____ each
(N)	Bead tie-in:		
	Heel ply turn-ups	(as determined)	_____ each
	Toe ply turn-ups	(as determined)	_____ each
	Flippers/Bead bundle	(as determined)	_____ each
(O)	Minimum load capacity of cord	(as determined)	_____ lbs
(P)	Cord count in crown	(as determined)	Plies___ Count____ Each
	Measured at 90º angle to cord axis		Plies___ Count____ Each
			Plies___ Count____ Each
(Q)	Total crown thickness	(as determined)	_____ in
(R)	Number of breaker plies	(as determined)	_____ each
(S)	Number of tread reinforcing plies	(as determined)	_____ each
(T)	Sidewall vents per sidewall	(as determined)	_____ each
(U)	Durometer hardness	(as determined)	_____ Shore
(V)	Balance pad adhesion	Min 8 lb tubeless	_____ lbs
		Min 1-1/2 lb tubetyle	_____ lbs

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A.5.1 CONTACT AREAS SECTION.

CONTACT AREAS

The contact areas for the applicable load and inflation conditions are summarized below in accordance with [OEM/military specification] and MIL-DTL-5041. Footprints are located in paragraph 4.3.1.3.

Percent of Rated Load	Actual Load (lbs)	Inflation (psi)	Gross Contact Area (inch^2)	Net Contact Area (inch^2)
40				
55				
70				
85				
100 (Rated)				
115				
130				
145				
Bottomed				

[**Size/Ply Rating/Speed Rating**]

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A6. DYNAMIC TEST REQUIREMENTS SECTION.

DYNAMIC TEST REQUIREMENTS
[**Loads and Inflation Pressures**]
[**Test Sequence Requirements**]
Tire #1:
Tire #2:
Tire #3:
[**Cycle Destinations (including references to LST curves)**]
Taxi - Take-off
Extended taxi - Take-off
Taxi – High-speed take-off
Landing – Taxi
Taxi-out – Turn
Taxi-in – Turn
Taxi – Aborted take-off, Stop, Taxi
Envelope compliance
[**Special Requirements**]
Cable bruise
Step load obstruction

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A7. DYNAMIC TEST RESULTS SECTION.

DYNAMIC TIRE #1		
Test completion date:		
Test specification:		
Size, Ply rating, Type:		
Serial number:		
Actual tire weight (lbs):		
Bead seating pressure (psi):		
Dimensions	After 12-hours minimum growth	
Rated inflation (psi)		
Outside diameter (inch)		
Section width (inch)		
Shoulder diameter (inch}		
Shoulder width (inch)		
Flat plate deflection at rated load/pressure	Before break-in	After Break-In
(inch)		
(%)		
Curved surface inflation before break-in (psi)		
[**Name of test cycle**] [LST Curve No. or Figure No.]	Taxi	Take-off
Flywheel outside diameter (inch)		
Test inflation (psi)		
Acceleration (ft/sec^2)		
Speed range (knots)		
Roll distance (ft)		
Taxi time (sec}		
Time programmed take-off (sec)		
Load range (lbs)		
Average peak carcass air temp (°F)		
Number of test cycles		
Remarks: [Tire condition following completion of test cycles shall be described in this section.]		

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A7. DYNAMIC TEST RESULTS SECTION. DYNAMIC TIRE #1 (Continued)

[**Name of Test Cycle**] [LST Curve No. or Figure No.]	LANDING	TAXI
Flywheel outside diameter (inch)		
Test inflation (psi)		
Deceleration rate (ft/sec^2)		
Speed range (knots)		
Roll distance (ft)		
Taxi time (sec)		
Time programmed landing (sec)		
Load range (lbs)		
Average peak carcass temp (°F)		
Number of test cycles		
Remarks: [Tire condition following completion of test cycles shall be described in section]		
[**Name of Test Cycle**] [LST Curve No. or Figure No.]	TAXI	TAKE-OFF
Flywheel outside diameter (inch)		
Test inflation (psi)		
Deceleration rate (ft/sec^2)		
Speed range (knots)		
Roll distance (ft)		
Taxi time (sec)		
Time programmed landing (sec)		
Load range (lbs)		
Average peak carcass temp (°F)		
Number of test cycles		
Remarks: [Summary of all test results shall be provided in this section. Any evidence of tread chunking or groove cracking shall be noted. Refer to submitted cut-section or photo as necessary.]		

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A8. OTHER PERTINENT DATA SECTION

Low Temperature Brittleness Qualification (Tubeless Tire Liner Compound Only)

Aircraft tire liner compound construction [#] has met the low temperature requirement in accordance with paragraphs 4.7.7 of MIL-DTL-5041. The brittleness temperature of construction [#], as defined by ASTM D746, is _____ degrees °F. A copy of the USAF approval letter is located in [appendix].

High Temperature Compliance

Quality Control

Preproduction acceptance criteria are established for control of raw material to be used in all aircraft tires (qualification test tires and production tires) manufactured at [facility location]. This then assures material used in production tires meets the same control limit as that used in the qualification tires. Dynamic testing assures these materials with the established limits are adequate.

Qualification test tires represented in this QTR were manufactured in the [name] facility under the same quality assurance criteria and test procedures that are applicable production tires.

Product Identification And Marking

The identification markings on the tire conform to section 3.6 of MIL-DTL-5041 [and to section _____of OEM/military specification].

Photograph Cross-Section

A photograph, which clearly depicts (adequate contrast) the cross-sectional design of the subject tire (i.e. number of tread plies, number of tread reinforcing plies, number of breaker plies, number of carcass plies, colored wear indicator), is located in [appendix].

Cut Limit

A cut limit of [X/32] is recommended.

Colored Wear Indicator (If Applicable)

In accordance with [OEM/military specification], a [color] colored wear indicator ply is located [just below or just above] the first carcass ply.

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – AS
Air Force – 70

Preparing Activity:
Air Force – 70

Review Activities:
United States Marine Corps – MC

Project 2620-2018-002

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